

Estimating the Impact of Amazon's Price War: Evidence from the Quidsi Acquisition

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This paper investigates how Amazon's 2010 price war with Diapers.com and subsequent acquisition of its parent company, Quidsi, altered consumer experiences in the diaper product market. Using a monthly panel of Amazon product categories from USCD's dataset, diapers as the treatment group, other baby products as the control group, we estimate the causal effect of Amazon's intervention on the average product ratings and review volume. After aggregating millions of product-level reviews spanning 1996-2023, we implemented a difference-in-differences (DiD) design using product-level treatment assignment and the November 2010 acquisition as the intervention date. The findings show that diaper products experienced a disproportionately large increase in average rating compared to control categories following the acquisition, suggesting that the quality of diapers on Amazon did not decline as a result of the acquisition, and that other possible factors like Amazon's pricing strategies, improved logistics, and platform visibility generated a category-specific improvement in consumer sentiment beyond broader marketplace trends. These results emphasize the role of platform power in shaping perceived product quality and carry implications for the nuances of competition policy and digital-market oversight.

Introduction

Price competition in e-commerce markets has intensified dramatically over the last two decades as dominant firms leverage scale, logistics, and algorithmic visibility to secure market advantage. While economic research traditionally focuses on how price wars affect market share and firm competition (Bernard, 2016), less research is done on how these conflicts affect consumer sentiment. Product reviews and ratings, which are now integral components of online shopping, provide a great opportunity to measure the downstream effects of large-scale platform interventions.

One of the clearest examples of aggressive platform-driven competition is Amazon’s price war with Diapers.com, whose parent company is Quidsi, in 2010. Diapers.com, founded in 2005, grew rapidly as a specialized e-commerce retailer of baby essentials. Amazon responded by launching steep price cuts in early 2010, effectively initiating a price war. Amazon ultimately acquired Quidsi in November 2010 and fully shut down its operations by 2017. This acquisition provides a unique opportunity to examine how platform behavior reshapes consumer experience and affects consumer sentiment. If Amazon’s intervention led to lower prices, improved shipping, or increased visibility for the diaper product, we would expect corresponding changes in visible consumer metrics, specifically product ratings and review volume. And if the intervention led to lower competition, higher prices, and less pressure to maintain quality, then we would expect ratings to decline, whether review count increased or not.

Previous research about price competition in regards to Amazon shows that reductions in competition lead to price increases for related products (He et al., 2022). When Toys’R’Us shut down, toys saw a quick 5% increase in price, especially in larger toys that were more likely to be stocked by Toys’R’Us. Our study investigates whether Amazon’s 2010 intervention caused a category-specific improvement in consumer sentiment towards baby products, with a focus on diaper products. Diapers are more likely to be affected by the acquisition of diapers.com as its namesake and highest-selling product. Using monthly panel data from the UCSD Amazon Review Dataset (1996-2023), we implemented a difference-in-differences (DiD) strategy comparing diaper products (treatment) with similar product categories less affected by the price war (control). We further complement the DiD analysis with product-level regressions that explore the determinants of individuals’ review ratings, providing insights into why ratings behave differently between diapers and non-diaper baby products.

We find strong evidence that Amazon’s price war and acquisition had a statistically significant positive effect on diaper product ratings. Diaper products show a distinct post-acquisition jump in both rating level and review volume, at a rate greater than non-diaper products. Additional regression results reveal meaningful structural differences between how different factors affect ratings of diaper and non-diaper products. Diaper ratings are more sensitive to popularity and pricing, and exhibit greater variability unexplained by observable product attributes. These findings emphasize the ways platform-driven interventions, not just intrinsic product quality, shape consumer perception in online marketplaces.

Data Description

The analysis uses the UCSD Amazon Review Dataset (1996-2023), a large-scale collection of Amazon review ratings (continuous 1-5), timestamps, and product metadata. Our data includes over 6 million reviews on Amazon labeled under ‘Baby Products’, including variables such as rating, timestamp, parent ASIN, helpful votes, number of ratings, and average rating. This enables the construction of aggregate monthly time series for each product family, regression models, and difference-in-difference models. The

original dataset separates the review data and product data, so we joined the information together using the ASIN or product ID found in the reviews.

1. Categorization of Product Families

Products are mapped into two groups: diaper products (treatment) and non-diaper baby products (control) based on the name/title of the product.

2. Monthly Aggregation

For each category, we computed: monthly average rating, monthly number of reviews, and total review activity.

3. Intervention Timing

- Pre-treatment period: January 1996 - November 7, 2010
- Post-treatment period: November 8, 2010 - 2023

The intervention date in our model is the day that the acquisition of Quidsi (diaper.com) was officially announced, which is November 8, 2010.

4. Outcome Variables

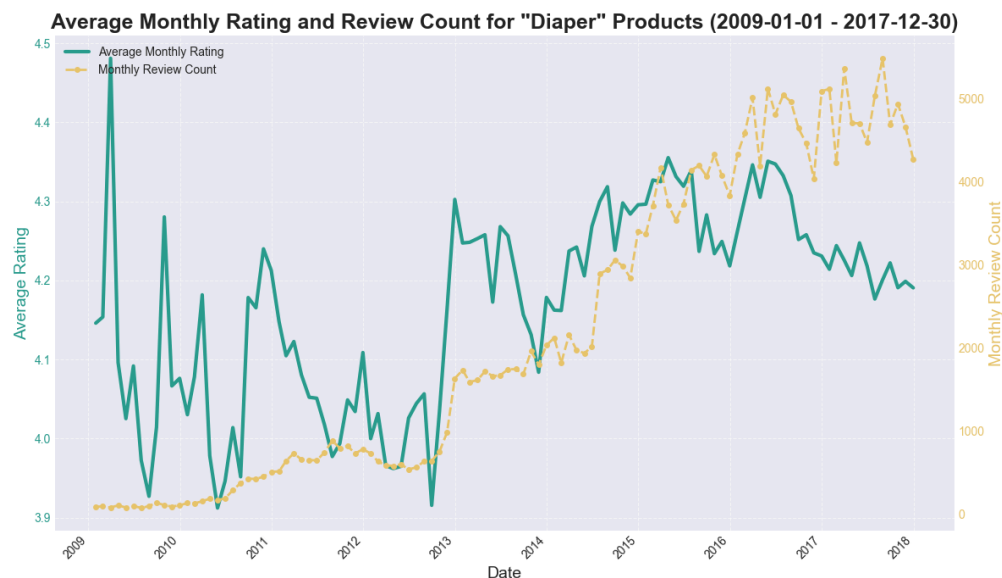
- Average Rating (continuous 1-5)
- Review Count (non-negative integer)

5. Final Dataset

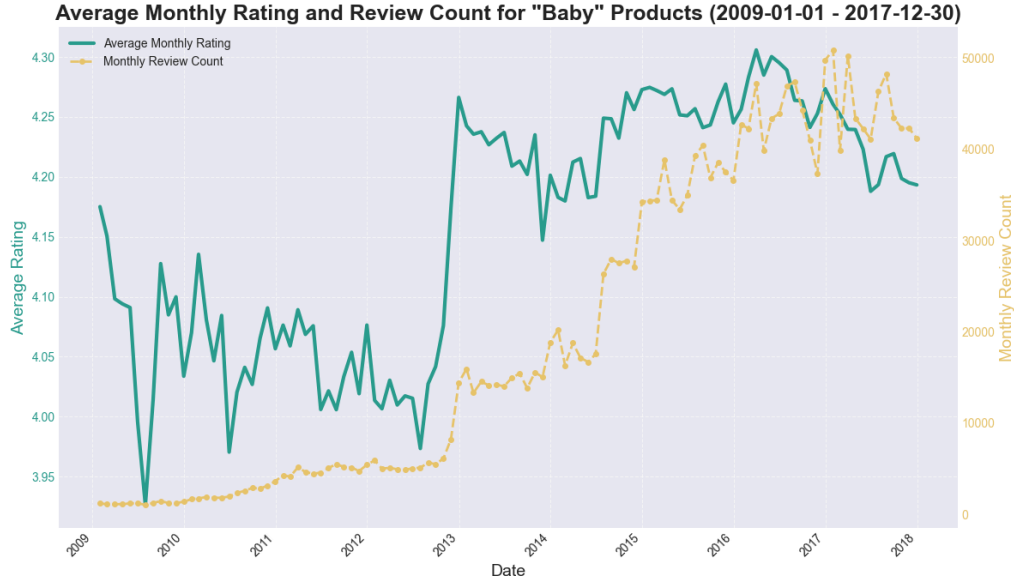
The full panel contains monthly aggregates for three categories over 27 years, resulting in hundreds of time-category observations.

Tools and Economic Models

We aggregate the reviews into a monthly rating and monthly count for both the diaper and non-diaper groups and plot them with a line graph. The left y-axis in blue represents the ratings, and the right y-axis in yellow represents the review counts. We limited the data for the line graph to a year before the start of the price war and the year after Amazon fully shut down Quidsi in order to better visualize changes during the timeframe.



Graph 1: Average Monthly Rating and Review Count for Diaper Products



Graph 2: Average Monthly Rating and Review Count for Non-Diaper Products

Difference in Difference Model:

$$rating_i = \beta_0 + \beta_1 Post_i + \beta_2 Treatment_i + \beta_3 Post_i * Treatment_i + e_i$$

We use a standard difference in difference model in order to measure the effects of the acquisition on the ratings of diaper products. Post is a binary variable representing reviews before the acquisition as 0, and reviews after the acquisition as 1. Treatment is a binary variable representing non-diaper product reviews as 0 and diaper product reviews as 1. The error term to capture variations unexplained by our model is represented by e_i . Using this model, we are able to estimate the change in diaper reviews as a result of the acquisition, while controlling for changes in the non-diaper/control group.

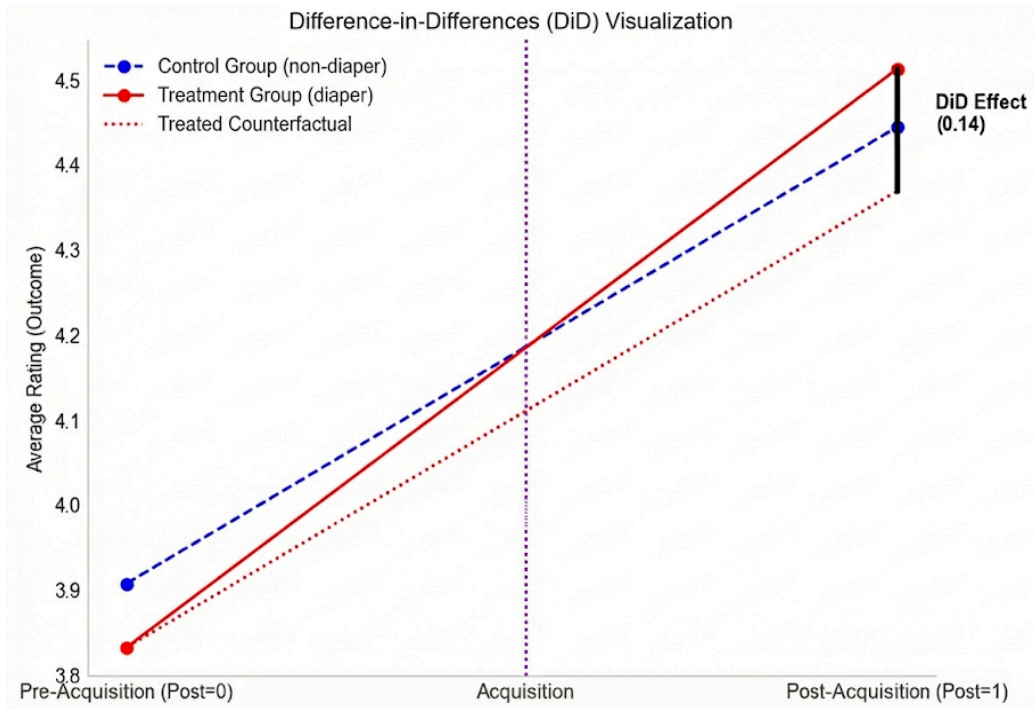
After creating the model from our Amazon reviews:

β_0 represents the average rating of non-diaper products before the acquisition.

$\beta_0 + \beta_1$ represents the average rating of non-diaper products after the acquisition.

$\beta_0 + \beta_2$ represents the average rating of diaper products before the acquisition.

$\beta_0 + \beta_1 + \beta_2 + \beta_3$ represents the average rating of diaper products after the acquisition, with the difference in difference effect measured as β_3 .



Graph 3: Difference in Difference graph

Table 1: Difference-in-Differences OLS Regression Results

Dependent Variable:	(1) Average Rating
Treated × Post	0.1436*** (0.005)
Post	0.5368*** (0.002)
Treated	-0.0752*** (0.005)
Constant	3.9073*** (0.002)
Observations	6,028,954
R-squared	0.026

Note: Standard errors are reported in parentheses below the coefficients. *** p<0.01, ** p<0.05, * p<0.1

Regression Model:

$$rating_i = \beta_0 + \beta_1 NumRatings_i + \beta_2 NumUpvotes_i + \beta_3 avgRating_i + \beta_4 Price_i + e_i$$

We use a standard Ordinary Least Squares (OLS) regression model in order to predict how different factors like the number of ratings, number of helpful upvotes, average rating of the product, and price of the product in 2023 affect the rating, while holding factors about the product constant, like price and average rating.

Table 2: OLS regression results

	(1)	(2)
	Baby Product	Baby Product
VARIABLES	Non-diaper rating	Diaper rating
Number of ratings	-2.28e-06*** (4.28e-08)	-3.68e-06*** (1.14e-07)
Number of helpful vote	-0.00125*** (0.000145)	-0.00276*** (0.000778)
Product's overall rating	1.205*** (0.00290)	0.917*** (0.0107)
Price (2023)	-0.000239*** (1.25e-05)	-0.00274*** (8.75e-05)
Constant	-1.194*** (0.0133)	0.147*** (0.0497)
Observations	3,529,199	364,019
Adjusted R-squared	0.064	0.029

Robust standard errors in parentheses
*** p<0.01, ** p<0.05, * p<0.1

Results

A. Difference-in-Difference Model Interpretation

From our difference-in-difference model, we find that the average rating of diapers before the acquisition was lower than the overall non-diaper products, but after the acquisition, the average rating of diaper products had a significant positive increase over the non-diaper products.

Specifically, on a 1-5 scale, diapers were on average 0.0752 points lower than the control before the acquisition, and 0.0684 points higher than the control after it, meaning the difference-in-difference effect was 0.1436 points. While we did expect a significant change after the acquisition, the expected ratings of diapers increased after the acquisition when we expected it to decrease, meaning that the acquisition did not decrease reviewers' perceptions of diaper and non-diaper baby products on Amazon.

B. Regression Table Interpretation

The regression table provides deeper insight into rating behavior:

- Product popularity (number of ratings) negatively predicts individual rating
 - Non-diaper: -2.28e-06 (p < 0.01)
 - Diaper: -3.68e-06 (p < 0.01)

Products with more reviews tend to receive slightly lower individual ratings. This effect is larger in magnitude for diapers, suggesting that popular diaper products receive more scrutiny, the diaper category has more heterogeneous buyers, and consumer expectations are stricter for essential goods.

2. Helpful votes also trend negative

Negative coefficients imply that more “helpful” reviews tend to be critical. In diapers, the coefficient is larger, meaning that for diaper and non-diaper ratings with similar helpfulness, diaper ratings will be more negative, holding the other factors constant.

3. The product’s overall rating is the strongest predictor

Both regressions show large positive and highly significant coefficients: 1.205 for non-diaper products and 0.917 for diaper products. This means that individuals' ratings tend to align with the product’s historical reputation. The lower coefficient for diapers suggests more variance among diaper reviews.

4. Higher prices correlate with lower ratings

- Non-diaper: -0.0002039 ($p < 0.01$)
- Diaper: -0.00274 ($p < 0.01$)

The effect of prices on rating is negative for both groups, but for diaper products, the effect of price is much stronger. Consumers purchasing diapers, an essential, frequently repurchased item, are more price-sensitive and more likely to penalize perceived overpricing.

5. Diaper review behavior is harder to predict

- Adjusted R-squared (non-diaper): 0.064
- Adjusted R-squared (diaper): 0.029

This suggests that ratings for diapers are more influenced by unobserved factors in our model (e.g., leakage incidents, packaging issues, delivery quality), and diapers have more rating variability not explained by basic product attributes.

Discussion of Findings

Our results indicate that Amazon’s intervention fundamentally altered consumer sentiment toward diaper products. The DiD analysis identifies a clear, causal improvement in diaper ratings following the 2010 acquisition, while the product-level regression reveals structural differences between diaper and non-diaper goods that help explain this shift. More specifically, consumers in the diaper market are more critical and more price-sensitive. While the decrease in competition leading to more positive consumer ratings is unexpected from a typical economic lens, we believe that this discrepancy is due to Amazon not being the fully-formed e-commerce giant it is now back in 2010. Amazon’s power in books had been established then, but it had not yet gained dominance in other product segments like baby products. In order to grow and maintain its growth in new products, Amazon is incentivized to make its offerings as attractive as possible in order to capture as much of the market as it can. If Amazon solely bought diapers.com to shut it down, substitutes in the form of brick-and-mortar stores or new e-commerce competition could have captured those consumers. Instead, Amazon is encouraged to make its product as good as possible, whether it be through superior logistics, pricing mechanisms to amplify perceived quality, or using its newly acquired market power to get better deals on price and quality from suppliers. Lastly, algorithmic visibility may have boosted perceived trustworthiness, as diaper listings gained more prominent placement post-acquisition. These dynamics illustrate how platform power can influence

marketplace outcomes beyond simple price competition, as they alter consumer perceptions, product reputations, and long-term demand patterns.

Conclusion

Amazon's price war and subsequent acquisition of Diapers.com had a measurable and statistically significant positive effect on diaper product ratings on its platform. By combining causal inference with product-level regression analysis, this study shows that diaper ratings increased after the 2010 intervention, review activity surged uniquely in the diaper category, and diaper review behavior is structurally different from other baby products; being more price-sensitive, more variable, and less predictable. The difference-in-difference effect strongly supports an interpretation that Amazon consumers benefited from Amazon's acquisition of Quidsi, or that Amazon's pricing and logistics changes improved consumer experiences, or at least consumer perceptions, within the diaper category. Together, these findings underscore the power of platform-driven interventions in shaping consumer perception and the benefits of acquisitions. As digital marketplaces become increasingly centralized, understanding the interplay between platform strategy, consumer sentiment, and market competition remains essential for policymakers, consumers, and the platforms themselves.

Limitations and Future Research

This study has several limitations that should be considered when interpreting the results. First, although the difference-in-differences framework helps isolate the effect of Amazon's acquisition of Quidsi, unobserved platform-level changes, such as shifts in search algorithms or marketing strategies, may also influence product ratings and review behavior. Second, the analysis relies on ratings and review counts as proxies for consumer satisfaction, which may be affected by selection bias, reviewer heterogeneity, or strategic review behavior rather than true changes in product quality. Third, the use of category-level aggregation may mask important variation across individual products within the diaper category, particularly if Amazon's intervention affected some brands or price segments more than others. Additionally, changes in the composition of reviewers over time, including the growth of Amazon Prime users, could contribute to shifts in ratings independent of the acquisition itself. Future research could address these limitations by conducting product-level difference-in-differences analysis, incorporating textual sentiment analysis of review content, or distinguishing between Prime and non-Prime reviewers to better understand the mechanisms driving changes in consumer evaluations. Further work examining how platform interventions affect different types of essential goods would also help generalize the findings beyond the diaper market.

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